FTransDeepLab: Multimodal Fusion Transformer-Based DeepLabv3+ for Remote Sensing Semantic Segmentation

Haixia Feng, Qingwu Hu, Pengcheng Zhao, Shunli Wang, Mingyao Ai, Daoyuan Zheng, and Tiancheng Liu

***Abstract*— High-resolution remote sensing images contain rich color and texture information, but due to the inherent limitations of 2-D data, achieving high-quality semantic segmentation remains a challenge. Multimodal data fusion technology has emerged as an effective approach to overcome this issue. To accurately capture the semantic information in remote sensing images, this study designs a multimodal fusion Transformer-based DeepLabv3+ model for remote sensing semantic segmentation, named FTransDeepLab. Specifically, the network learns features from two modalities and is inspired by the DeepLab architecture. We extended the encoder by stacking the multiscale Segformer, encoding the input images into highly representative spatial features. Additionally, we introduced the multimodal feature rectification (MFR) module and the multimodal feature fusion (MFF) module. The MFR, composed of a channel attention module and a spatial attention module, enhances the model's ability to capture essential features and improves performance by focusing on both global and local contexts. The MFF module utilizes a cross-attention mechanism to optimize the feature fusion process, which enhances representation learning by facilitating the interaction between diverse information and integrates features from different modalities. Finally, in the decoding path, the extracted high-level features are concatenated with low-level features to optimize the feature representation and upsampled to restore the size of input image. Extensive results on two datasets, the International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen and Potsdam, have confirmed that the proposed FTransDeepLab can achieve superior performance compared to the state-of-the-art segmentation methods.**

***Index Terms*— DeepLabv3+, multimodal fusion, remote sensing, semantic segmentation, vision transformer (ViT).**

I. Introduction

SEMANTIC segmentation plays a crucial role in the intelligent interpretation of remote sensing and computer vision (CV) [1]. It involves extracting semantic information from high-dimensional feature maps and predicting objects such as car, building, road, and tree at the pixel level with dense accuracy [2], [3], [4]. Before the advent of deep learning, traditional image segmentation methods, typically based on thresholding, contour, and region information, rely on manual modeling and tuning to obtain segmentation results. Although these methods are efficient, they lack accuracy [5]. In contrast, deep learning technologies, particularly convolutional neural network (CNN) [6], have emerged as the vanguard in image fusion-based semantic segmentation studies. Later, Dosovitskiy [7] and Islam [8] apply the Transformer to the CV, proposing the vision transformer (ViT) architecture. Unlike CNN structures, ViT models long-distance dependencies between pixels through self-attention mechanisms, thus excavating high-level semantic features [9]. Theoretically, ViT has a global receptive field for the input vector at the feature extraction stage, truly breaking through the locality of convolution operations. Thus, this mode is also a new pattern of feature extraction. Additionally, ViT's ability to mine big data surpasses that of CNNs, aligning well with today's era of big data. ViT variants are mostly based on the Unet structure, enhancing segmentation accuracy by integrating Transformer, which has also achieved remarkable success [10], [11], [12], [13], [14]. Many studies have attempted to combine Unet with Transformer models, such as Trans-Unet [15] and Swin-Unet [16].

Unlike natural images, high-resolution remote sensing images typically have larger image sizes, cover vast areas, and feature a wide variety of objects [17]. Therefore, semantic segmentation usually requires rich semantic representations and contextual information to improve the model's recognition accuracy. When a single-modal data source is insufficient for information extraction, multimodal data fusion strategies are often employed to enhance prediction accuracy [18]. Remote sensing data present a variety of rich data modes beyond red-green-blue (RGB) images. For example, RGB-depth [19] where depth information helps in identifying object boundaries and RGB and light detection and ranging (RGB-LiDAR) [20] provides point cloud information for quickly obtaining high-precision, high-resolution elevation information of the terrain. This fusion enhances the complementary advantages of spectral and elevation information, benefiting semantic segmentation of remote sensing images [21], [22]. Research has shown that using 3-D height data can improve the performance of semantic segmentation in remote sensing data. Different types of land cover generally have unique height attributes, making the geometrical information in height data highly correlated with the semantic information in remote sensing imagery [23]. Regarding multimodal fusion in remote sensing, researchers have explored four different fusion strategies [24]:

Received 1 December 2024; revised 12 February 2025 and 12 March 2025; accepted 18 March 2025. Date of publication 21 March 2025; date of current version 4 April 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFB2600401; and in part by the Key Laboratory of Land Satellite Remote Sensing Application, Ministry of Natural Resources of China, under Grant KLSMNR-G202221. *(Corresponding author: Qingwu Hu.)*

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Digital Object Identifier 10.1109/TGRS.2025.3553478

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early fusion, intermediate fusion, late fusion, and multilevel fusion. In general, early fusion [25], [26] requires proper alignment of multimodal data and may lack robustness to task-irrelevant information, while late fusion [27], [28] is limited in its ability to leverage interdependencies between multimodal data. In contrast, intermediate fusion [29], [30] can capture cross-modal dependencies in feature representations, making it more effective in the context of representation learning. Multilevel fusion [31], [32] performs feature fusion at multiple levels, integrating complementary depth information into the semantic segmentation framework.

Compared to Unet [33], the DeepLabv3+ [34] demonstrates outstanding performance in segmentation tasks. Inspired by this, this article proposes a multimodal mid-level fusion architecture similar to DeepLab, termed multimodal fusion Transformer-based DeepLabv3+ model (FTransDeepLab), to effectively capture long-range dependencies and multiscale features in multimodal data. Specifically, to model the atrous spatial pyramid pooling (ASPP), this work first designs a stacking of the multiscale Segformer [35] to encode input images into highly representative spatial features. Subsequently, the obtained features are input into a multimodal feature rectification (MFR) module composed of a channel attention module and a spatial attention module, which enhances the model's ability to capture essential features and improves performance by focusing on both global and local contexts. The core of the channel attention module lies in utilizing the channel relationships between multisource features to generate multiple channel attention maps, focusing on the learning of global weights for comprehensive refinement. In contrast to the channel attention that emphasizes global refinement, the spatial attention module focuses more on the learning of local weights, allowing for fine-grained local refinement. We adopt a cross-attention mechanism to optimize the feature fusion process, named multimodal feature fusion (MFF), which effectively enhances representation learning by facilitating the interaction between diverse information and integrates features from different modalities. Using a cross-attention mechanism, attention weights are calculated between the refined multimodal feature sequences, followed by weighted connection and embedding back to the original size. Finally, in the decoder, the extracted high-level features undergo bilinear upsampling and are then concatenated with low-level features from the encoder to optimize feature representation, using upsampling to restore the input image size. The contributions of this work are threefold, as summarized in the following.

1) A multimodal fusion Transformer-based DeepLabv3+ model for remote sensing semantic segmentation has been developed. The proposed network, named FTransDeepLab, integrates the advantages of hierarchical Segformer into DeepLab's encoder, allowing FTransDeepLab to effectively capture long-range and multiscale representations.
2) The introduction of an MFR and MFF. The MFR, composed of a channel attention module and a spatial attention module, enhances the model's ability to capture essential features and improves performance by focusing on both global and local contexts. The MFF, which employs a cross-attention mechanism, enhances representation learning by facilitating the interaction between diverse information sources and integrates features from different modalities.
3) Extensive experiments on two fine-resolution remote sensing datasets, International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen and Potsdam, confirm that the proposed FTransDeepLab substantially outperforms existing models.

The remainder of this article is organized as follows. Section II first reviews related work on semantic segmentation methods and multimodal data fusion. Then, Section III presents the structure of the proposed FTransDeepLab, while Section IV provides detailed information about the extensive experiments conducted. Section V discusses a comprehensive analysis of the model. Finally, conclusions are given in Section VI.

II. Related Work

A. Multimodal Data Fusion

Multimodal data fusion is becoming increasingly popular in the fields of CV and remote sensing. Multimodal semantic segmentation methods are considered effective for achieving more accurate segmentation by extracting information from source images of different modalities, allowing for richer feature representations compared to single-modal images. In remote sensing image fusion, multisource data fusion combines high spatial resolution panchromatic images with infrared-red-green (IRRG), depth information, point cloud information, spectral information, elevation information, thermal information, and more. Research on RGB and other modalities for semantic segmentation has been ongoing for nearly a decade.

A simple approach to multimodal information fusion is to directly concatenate additional modality information with the RGB. For example, Gupta et al. [36] fused RGB and depth information by adding depth images as an additional channel to RGB images during the input stage. Additionally, other modality information can be input as a second branch. Hazirbas et al. [37] designed a straightforward dual-branch network architecture to fuse RGB and digital surface model (DSM) data. Specifically, during the encoding stage, each branch processes RGB and DSM data in parallel, and the feature maps generated by the two branches are added elementwise after each convolutional block. While this direct approach improves segmentation accuracy, it does not fully leverage the unique features of each modality.

Subsequent research has explored multistage fusion architectures to improve model accuracy. For instance, Hu et al. [38] proposed an attention-complementary network that fuses RGB and depth information at multiple stages. Chen et al. [39] proposed GLPNet, which uses a local context fusion module (L-CFM) for dynamic alignment and a global context fusion module (G-CFM) to effectively propagate depth information to the RGB branch, jointly modeling multimodal global context features. Zhou et al. [40] proposed an efficient spatial attention network, designing dual branches for RGB and

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depth information fusion. Wang et al. [41] introduced the token fusion model for RGB and depth semantic segmentation and 3-D object detection by combining point cloud and image data. Li et al. [65] proposed a dual-stream 2-D CNN architecture that extracts spectral and spatial features and adaptively learns the fusion weights of spectral–spatial features, enabling efficient hyperspectral image classification. Yang et al. [66] introduced an unsupervised deep tensor network (UDTN) that combines tensor theory with deep learning to achieve effective hyperspectral and multispectral image fusion. This method incorporates tensor filtering layers and a coupled tensor filtering module, leveraging a shared encoding tensor to simultaneously encode spectral and spatial features, and is trained in an unsupervised end-to-end manner. Wan et al. [42] introduced a Mamba-based fusion mechanism to enhance the model's channelwise modeling ability, effectively selecting critical information from RGB and depth data. Ma et al. [43] developed a feature fusion hypergraph neural network for hyperspectral image classification, effectively combining CNN features, spectral features, and spatial features. He et al. [44] integrated CNN and Transformer in a model to fuse RGB and elevation information, achieving a lightweight multimodal semantic segmentation structure. He et al. [45] introduced a modality difference adaptive fusion module and a spatial semantic fusion module to adaptively fuse RGB and thermal information, capturing multiscale perceptive fields and integrating shallow-level features rich in visual information with deep-level features strong in semantics. Wang et al. [46] proposed a semantic-guided fusion network consisting of an asymmetric encoder for RGB and thermal information and a decoder, enhancing fused features with cross-layer and semantic information. Xue et al. [67] proposed a high-dimensional image representation framework based on a tensor convolution-like low-rank dictionary (TCLRD). By employing convolution-like low-rank dictionary learning and coefficient coding, this framework captures the low-dimensional shift-invariant characteristics of tensor data and is applied to tasks such as image denoising, hyperspectral image completion, and image clustering. Wang et al. [47] proposed a multimodal-aware semantic segmentation model, leveraging channel attention to focus on high-frequency information and utilizing three or more modalities to address multimodal semantic segmentation for modality-agnostic cases.

B. Semantic Segmentation of Remote Sensing

Compared with natural images [48], remote sensing images typically contain more diverse and complex features. Semantic segmentation, a technique for labeling each pixel to differentiate objects, is often achieved through statistical analysis, handcrafted feature-aided learning, or data-driven deep learning methods. Traditional remote sensing semantic segmentation methods [48], [49] primarily extract low-level and handcrafted features from input images, which are then mapped to label categories using supervised classifiers.

In recent years, advanced remote sensing semantic segmentation methods have adopted encoder–decoder architectures, known for their stable and reliable feature extraction and transformation capabilities. Unet has been extensively studied and applied in remote sensing images. TreeUNet [50] introduced a Tree-CNN module, which adaptively improves pixel-level classification accuracy by fusing multiscale features through a confusion matrix. Meanwhile, ResUNet-a [51] extended the Unet baseline by incorporating residual connections, atrous convolutions, pyramid scene parsing, and multitask inference.

To address challenges such as scale variation and object diversity in remote sensing images, many studies have proposed novel solutions. For instance, Zheng et al. [52] enhanced the recognition of foreground features by learning contextual information related to foreground modeling. Ma et al. [53] proposed a foreground-activated object representation framework to strengthen weak features in small objects. Wang et al. [54] introduced the Swin Transformer as a backbone network to extract contextual information and designed a novel dense connected feature aggregation module (DCFAM) decoder to restore resolution and generate segmentation maps. Kang et al. [55] considered relationships between different objects (e.g., semantics, shapes, and materials) and proposed a hierarchical class graph for semantic segmentation of high-resolution remote sensing images to learn structured relationships among various objects. Peng et al. [56] precisely encoded small-scale semantic information by selecting informative features and expanding the receptive field of low-level feature maps, significantly improving the accuracy of small object segmentation. Jin et al. [57] proposed a multimodal fusion network based on edge detection, using boundary spatial information to assist multimodal information fusion. Li et al. [58] introduced an edge detection branch stream (EDBS) with a composite loss function to compensate for boundary information loss in the semantic segmentation branch stream (SSBS).

Another key research focus has been improving segmentation results using attention mechanisms. Xiao et al. [59] combined the advantages of convolution operations and Transformers, introducing a deformable self-attention mechanism to automatically adjust the receptive field and enhance multiscale representation learning. Zhao et al. [60] incorporated attention mechanisms in the ASPP module, achieving adaptive feature refinement by fusing pooling index maps and high-level feature maps, thereby recovering fine-grained features. Ma et al. [61] proposed FTransUNet, a multilevel multimodal fusion framework that combines CNN and ViT, introducing self-attention and cross-attention mechanisms to accurately represent local details and global semantics.

In the context of remote sensing image semantic segmentation, spatial details and channel information are crucial for highlighting differences between diverse objects. To effectively capture long-range dependencies and multiscale features across modalities, this article proposes a mid-level multimodal fusion architecture similar to DeepLab, termed the multimodal fusion Transformer-based DeepLabv3+ model (FTransDeepLab), incorporating MFR and MFF modules to capture interactions between modalities.

III. Method

Fig. 1 provides an overview of the proposed framework. FTransDeepLab is a network based on an encoder–decoder

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framework. We will specifically introduce the overall network structure of FTransDeepLab, as well as three key components in the networks encoder: Stacked Segformer, MFR, and MFF.

A. Overview of FTransDeepLab

In this section, we propose a multimodal fusion Transformer-based DeepLabv3+ model for remote sensing semantic segmentation. The network, named FTransDeepLab, is inspired by the DeepLab architecture. Specifically, given a remote sensing image $I_{\text{IRRG}} \in R^{C\times H\times W}$ and a normalized DSM (nDSM) image $I_{\text{nDSM}} \in R^{C\times H\times W}$ as the network inputs, where C, W, and W represent the number of channels, height, and width of the image, respectively, our network architecture has two branches in the input stage: the IRRG branch and the nDSM branch. At each stage, the data are fed into Segformer modules of different sizes for semantic feature extraction, reducing the height and width of the semantic feature maps while increasing the number of channels. The extracted features are then passed into the MFR module, which interacts to learn the interdependencies and complementary contextual information between the two modalities. These features are then passed to the MFF module, where cross-attention mechanisms are used to compute attention weights between the two different feature sequences for feature fusion. Finally, following the original DeepLabv3+ architecture, the extracted high-level features are upsampled using bilinear interpolation and concatenated with low-level features from the encoder to optimize the feature representation, followed by further upsampling to restore the size of input image.

B. Stacked Segformer

Inspired by Segformer's efficiency and ability to model long-range dependencies, we incorporated Stacked Segformer modules in the encoder, enhancing long-range information capture and improving segmentation of diverse object sizes in remote sensing images. As shown in Fig. 2, the network structure is designed with four stages, each consisting of an overlap patch embedding layer and multiple Transformer blocks, with MFR and MFF modules introduced after each stage. Upon receiving the input remote sensing image, the first stage extracts shallow features through the overlap patch embedding layer and three Transformer blocks. These features are then optimized using the MFR module and output through the MFF module. These shallow features mainly capture low-level semantic information, such as edges, textures, and color features. In the fourth and final stages, deep features are extracted through the overlap patch embedding layer and three Transformer blocks, optimized using the MFR module, and output through the MFF module. These high-level features contain rich global semantic information, enabling the model to better distinguish complex remote sensing scenes and object categories.

The overlap patch embedding layer contains a convolution layer, a flattening layer, and a layer normalization module. This process involves splitting the image and embedding it into a specific dimensional space, then flattening it along the height and width dimensions, and forming a 2-D matrix composed of a number of vectors and embedding dimensions, which is then used as the input for the Transformer block layer. The overlap patch embedding layer is configured with different parameters.

For an input image with dimensions $H \times W \times C$, in the ith stage of Transformer, multilayer images are obtained through overlap patch embedding, with corresponding resolution sizes $(H/2^{i+1}) \times (W/2^{i+1}) \times C_i$, where $i \in \{1, 2, 3, 4\}$, and the values of C_1, C_2, C_3, and C_4 are 64, 128, 320, and 512, respectively. After each stage, the height and width of the semantic feature map are halved, and the number of channels is progressively increased to enhance feature representation. Based on the characteristics of remote sensing images, we define K, E, S, and P, where K is the slice size, E is the embedding dimension size, S is the stride between two adjacent slices, and P is the padding size. In the first stage, $K = 7$, $E = 64$, $S = 4$, and $P = 3$ are set to perform the overlap patch embedding module, and in the second, third, and fourth stages, $K = 3$, $E = 128/320/512$, $S = 2$, and $P = 1$ are set to perform the overlap patch embedding module.

C. Multimodal Feature Rectification

To ensure efficient feature rectification before feature fusion, this study designs and introduces the MFR module, as shown in Fig. 3. This module consists of the channel attention module and spatial attention module, which focus on global and local contextual information, respectively. Together, they effectively enhance the model's ability to understand multimodal features and capture critical information, thereby significantly improving performance. Among them, the channel attention module plays a pivotal role in feature optimization and fusion efficiency. For the input intermediate feature maps $\mathbf{IF}_{\mathbf{IRRG}} \in R^{C\times H\times W}$ from IRRG and $\mathbf{IF}_{\mathbf{nDSM}} \in R^{C\times H\times W}$ from nDSM, the MFR module sequentially infers the channel attention vectors $\boldsymbol{W}^{\boldsymbol{C}}_{\mathbf{IRRG}} \in R^{1\times 1\times C}$ and $\boldsymbol{W}^{\boldsymbol{C}}_{\mathbf{nDSM}} \in R^{1\times 1\times C}$, and the spatial attention vectors $\boldsymbol{W}^{\boldsymbol{S}}_{\mathbf{IRRG}} \in R^{H\times W\times 1}$ and $\boldsymbol{W}^{\boldsymbol{S}}_{\mathbf{nDSM}} \in$. The overall channel and spatial rectification processes can be summarized as follows:

$$\begin{aligned}\mathbf{RF}_{\mathbf{IRRG}} &= \mathbf{IF}_{\mathbf{IRRG}} + \lambda_C\left(\boldsymbol{W}^{\boldsymbol{C}}_{\mathbf{nDSM}} \odot \mathbf{IF}_{\mathbf{nDSM}}\right) \\ &\quad + \lambda_S\left(\boldsymbol{W}^{\boldsymbol{S}}_{\mathbf{nDSM}} \odot \mathbf{IF}_{\mathbf{nDSM}}\right) \qquad (1)\\ \mathbf{RF}_{\mathbf{nDSM}} &= \mathbf{IF}_{\mathbf{nDSM}} + \lambda_C\left(\boldsymbol{W}^{\boldsymbol{C}}_{\mathbf{IRRG}} \odot \mathbf{IF}_{\mathbf{IRRG}}\right) \\ &\quad + \lambda_S\left(\boldsymbol{W}^{\boldsymbol{S}}_{\mathbf{IRRG}} \odot \mathbf{IF}_{\mathbf{IRRG}}\right) \qquad (2)\end{aligned}$$

where $\odot$ represents the elementwise multiplication. $\mathbf{RF}_{\mathbf{IRRG}}$ and $\mathbf{RF}_{\mathbf{nDSM}}$ are the rectified feature maps, and λ_C and λ_S are two hyperparameters, both with default values of 0.5.

The channel attention module's main task is to dynamically adjust the importance of each channel by analyzing the channel relationships between multimodal features, thereby generating multiple channel attention maps. These attention maps assign weights to different channels, effectively optimizing the global representation capability of the features. This ensures that key global features are highlighted while suppressing redundant or noisy information. This mechanism ensures effective complementarity of data from different modalities in the channel dimension, laying a solid foundation for subsequent feature fusion. The channel attention module achieves this by

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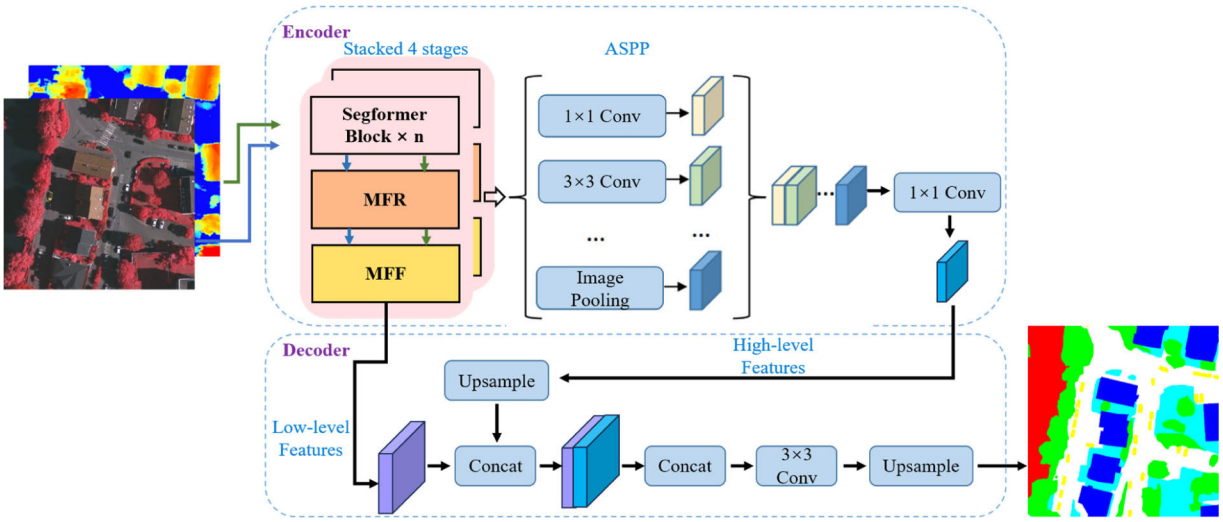


Fig. 1. Overview of the proposed FTransDeepLab. In this study, IRRG and nDSM are fed into the Segformer blocks. Subsequently, the obtained features are input into the MFR and MFF modules to enhance the model's capability and optimize the feature fusion process. Finally, the extracted high-level features are concatenated with low-level features and upsampled to restore the original input image size.

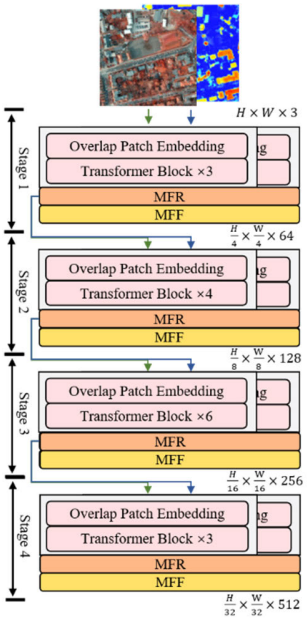


Fig. 2. Structure of stacked Segformer module with the overlap patch embedding layer and the Transformer block layer. For an input image with dimensions $H \times W \times C$, in the ith stage of Transformer, multilayer images are obtained through overlap patch embedding, with corresponding resolution sizes $(H/2^{i+1}) \times (W/2^{i+1}) \times C_i$, where $i \in \{1, 2, 3, 4\}$, and the values of C_1, C_2, C_3, and C_4 are 64, 128, 320, and 512, respectively.

leveraging a spatial dimension compression strategy to extract key features, generate global weights, and dynamically adjust channel importance. Through the collaborative optimization of multimodal features, it enables efficient fusion of global information, enhancing feature representation quality and model performance. The efficiency of the channel attention module partially comes from its design of the spatial dimension compression strategy. By compressing the 2-D spatial information into a 1-D channel dimension, the model focuses on channel relationships while reducing computational complexity and preserving critical global information. Specifically, the channel attention module employs both average pooling and max pooling to aggregate spatial information. Average pooling extracts the overall trend of the feature map, reflecting the importance of global features, while max pooling extracts salient region information, emphasizing the roles of critical channels. By combining average pooling and max pooling, the resulting feature vectors are concatenated into a fusion vector $\boldsymbol{V}^{\boldsymbol{C}} \in R^{1\times1\times4C}$. This fusion vector contains both global background information and salient local features, providing a rich source of information for calculating channel weights. The concatenated fusion vector $\boldsymbol{V}^{\boldsymbol{C}}$ is further input into a shared network. In the channel attention mechanism, the shared network refers to a multilayer perceptron (MLP) that processes the statistical information of all channels. The weights are initially randomly initialized and updated through optimization during training. The MLP models channel dependencies efficiently by performing scale transformations on the input: first compressing the concatenated channel dimension from $2C$ to C, extracting compact feature representations, and then expanding the channel dimension back to $2C$ to generate final attention weights. This compression and expansion process enables the model to efficiently capture both global and local channel relationships and generate normalized attention weights. Finally, the channel attention weights are split into two independent weight vectors $\boldsymbol{W}^{\boldsymbol{C}}_{\mathbf{IRRG}} \in R^{1\times1\times C}$ and $\boldsymbol{W}^{\boldsymbol{C}}_{\mathbf{nDSM}} \in R^{1\times1\times C}$, which are applied to the IRRG and nDSM feature maps, respectively. By weighting each channel, the channel attention dynamically adjusts the strength of features from different modalities, allowing the model to better balance the information distribution between multimodal features. For example, the color and texture information in IRRG data can be further highlighted through channel weighting, while the geometric and depth information in nDSM data can also be enhanced in specific channels. This collaborative optimization mechanism allows the model to integrate the advantages of multimodal data during fusion, significantly improving the quality of feature representation and the efficiency of fusion. The calculation process for the channel attention is as follows:

$$\boldsymbol{W}^{\boldsymbol{C}}_{\mathbf{IRRG}}, \boldsymbol{W}^{\boldsymbol{C}}_{\mathbf{nDSM}} = \mathrm{Split}\big(\mathrm{Sigmoid}\big(\mathrm{MLP}\big(\boldsymbol{V}^{\boldsymbol{C}}\big)\big)\big) \tag{3}$$

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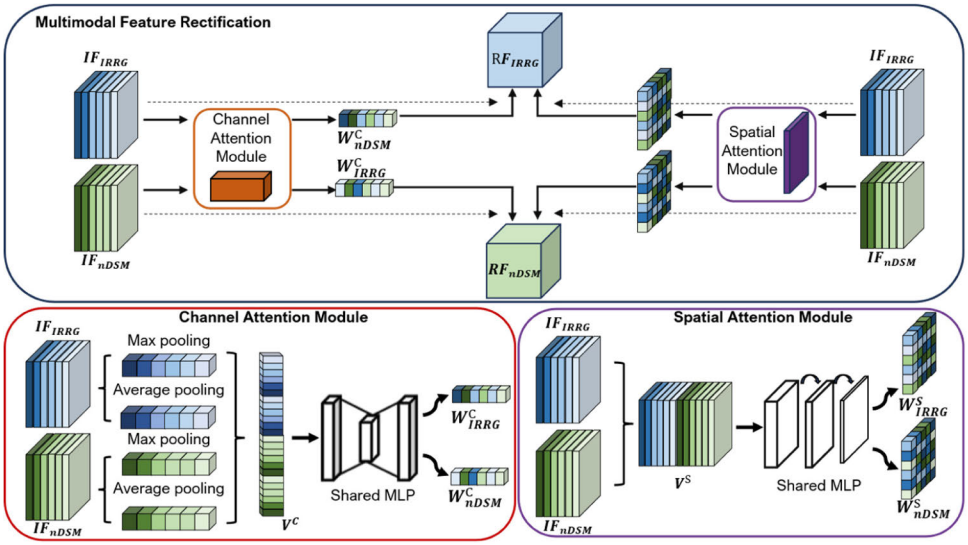


Fig. 3. Structure of MFR. This module is composed of a channel attention module and a spatial attention module. In the channel attention module, we combine average pooling and max pooling to aggregate the spatial information from the IRRG and nDSM feature maps. In the spatial attention module, we concatenate the IRRG and nDSM feature maps and input them into a shared network to generate spatial attention vectors.

$$V^C = \text{Concat}(\text{MaxPool}(\mathbf{IF_{IRRG}}), \text{AvgPool}(\mathbf{IF_{IRRG}}) \\ \text{MaxPool}(\mathbf{IF_{nDSM}}), \text{AvgPool}(\mathbf{IF_{IRRG}})). \quad (4)$$

The spatial attention module focuses on learning the weight distribution of local regions within the feature map, thereby achieving fine-grained local optimization. Unlike the channel attention module, which emphasizes the correction and optimization of global features, the spatial attention module dynamically adjusts the weight of each spatial position in the feature map. By highlighting the expressive power of key regions, it enhances the model's ability to understand complex spatial relationships. Before feature fusion, the spatial attention module refines and enhances features through meticulous correction and augmentation mechanisms, providing more precise input features for downstream tasks. The core mechanisms of the spatial attention module include feature concatenation and weight generation and allocation. First is the feature concatenation. The spatial attention computation process begins by concatenating the IRRG and nDSM feature maps along the channel dimension to form a fusion vector $V^S \in R^{H\times W\times 2C}$. This operation integrates multimodal features into a unified representation, laying the foundation for the subsequent generation of attention weights. Next is the weight generation and allocation. The concatenated fusion vector V^S is input into a shared network. In the spatial attention mechanism, the shared network is implemented through spatially shared convolution kernels, meaning that all spatial positions are processed using the same set of convolution kernels. The weights of the convolution kernels are randomly initialized and optimized during training via backpropagation. To accommodate the complexity of input features, the convolution kernels perform scale transformations on the spatial information: first, the concatenated input channel dimension is compressed from $2C$ to C; then, the channel dimension is expanded to 2 to generate the spatial attention maps corresponding to the two input feature maps. In this process, the model not only extracts global spatial information, but also uses the shared convolution kernels to generate weight maps that consistently reflect the importance of each spatial position, thereby improving feature representation. Finally, these attention maps are split into $\boldsymbol{W}^S_{\mathbf{IRRG}} \in R^{H\times W\times 1}$ and $\boldsymbol{W}^S_{\mathbf{nDSM}} \in R^{H\times W\times 1}$, which are used to dynamically adjust the importance of each spatial position in the IRRG and nDSM feature maps. The spatial attention computation process can be summarized as follows:

$$\boldsymbol{W}^S_{\mathbf{IRRG}}, \boldsymbol{W}^S_{\mathbf{nDSM}} = \text{Split}\big(\text{Sigmoid}\big(\text{MLP}\big(V^S\big)\big)\big) \quad (5)$$

$$V^S = \text{Concat}(\mathbf{IF_{IRRG}}, \mathbf{IF_{nDSM}}). \quad (6)$$

The channel and spatial attention mechanisms complement each other in feature correction: the channel attention provides a global optimization framework to ensure full interaction of global features from different modalities, while the spatial attention further refines the correction of local regions, making key features more prominent. Through the interactive application of weights, the IRRG and nDSM data achieve bidirectional complementarity and reinforcement in the channel and spatial dimensions. This collaborative correction method significantly improves the efficiency of feature fusion. On the one hand, the corrected IRRG data can more accurately express the spatial structure of targets; on the other hand, the corrected nDSM data acquire richer texture and color information. Ultimately, the MFR module, through the dual optimization mechanisms of global and local features, provides higher quality inputs for feature fusion, significantly enhancing the model's ability to parse and integrate multimodal data.

D. Multimodal Feature Fusion

Traditional feature fusion methods often rely on simple linear operations (such as weighted summation or direct concatenation). However, these methods exhibit significant limitations when dealing with complex data relationships and capturing deep interactions between modalities, making it difficult to meet the requirements of multimodal tasks.

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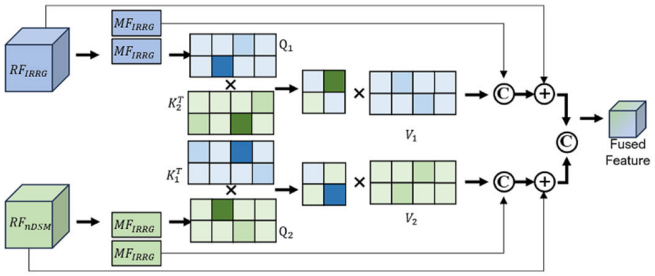


Fig. 4. Structure of MFF with cross-attention mechanism to optimize the feature fusion process.

To address this issue, this study proposes a cross-attention-based MFF module, as illustrated in Fig. 4. The MFF module dynamically optimizes the fusion process through explicit interactions between cross-modal features. This not only enhances the model's representation learning capabilities, but also significantly improves the comprehensive utilization of multimodal data. The core idea of the cross-attention mechanism is to calculate attention weights between feature sequences from two different modalities, deeply exploring semantic associations between modalities and dynamically adjusting the interaction strength between features. This mechanism effectively refines feature representations, highlights key information, and suppresses redundant features, thereby improving fusion efficiency. Compared to traditional methods, the cross-attention mechanism processes the complex relationships between multimodal data in a more flexible and precise manner, injecting higher expressive capabilities into the feature fusion process.

In the MFF module, the cross-attention mechanism first receives the corrected IRRG features $\mathbf{RF_{IRRG}} \in R^{C\times H\times W}$ and $\mathbf{RF_{nDSM}} \in R^{C\times H\times W}$. These features are flattened to a 2-D shape ($N = H \times W$) and linearly embedded into intermediate representations $\mathbf{MF_{IRRG}} \in R^{N\times C_i}$ and $\mathbf{MF_{nDSM}} \in R^{N\times C_i}$. Next, these features are further linearly mapped into query (Q), key (K), and value (V) matrices to prepare for cross-attention computation. By computing the query, key, and value matrices, the cross-attention mechanism can refine feature representations, highlight critical regions, and reduce redundant information. This refinement process enhances the semantic expressiveness of the fused features. The cross-attention computation process is the key to the module. For the IRRG modality, the query matrix $\boldsymbol{Q_1}$ and the key matrix $\boldsymbol{K_2}$ from the nDSM modality are multiplied to generate the cross-modal weight. Subsequently, the weights are multiplied with the value matrix $\boldsymbol{V_1}$, embedding important information from the nDSM modality into the IRRG representation. Similarly, for the nDSM modality, the critical information from the IRRG modality is effectively integrated into the nDSM representation. The above process can be expressed using the following formulas:

$$\mathrm{Cro}_1 = \mathrm{Softmax}\left(\frac{\boldsymbol{Q_1}\boldsymbol{K}_2^{\mathrm{T}}}{\sqrt{d}}\right)\boldsymbol{V_2} \tag{7}$$

$$\mathrm{Cro}_2 = \mathrm{Softmax}\left(\frac{\boldsymbol{Q_2}\boldsymbol{K}_1^{\mathrm{T}}}{\sqrt{d}}\right)\boldsymbol{V_1}. \tag{8}$$

After cross-modal attention computation, the results are concatenated with the features $\mathbf{MF_{IRRG}}$ and $\mathbf{MF_{nDSM}}$, enhanced intermediate feature representations. These concatenated features are linearly projected back to the same dimension as the original features ($N \times C_i$), preserving the contextual consistency of modality features while integrating cross-modal interaction information. Subsequently, these processed features are added back to the original features $\mathbf{RF_{IRRG}}$ and $\mathbf{RF_{nDSM}}$, generating the corrected output features

$$\mathbf{OF_{IRRG}} = \mathbf{RF_{IRRG}} + \mathrm{Proj}(\mathrm{Concat}(\mathbf{MF_{IRRG}}, \mathrm{Cro}_1(\mathbf{MF_{IRRG}}, \mathbf{MF_{nDSM}}))) \tag{9}$$

$$\mathbf{OF_{nDSM}} = \mathbf{RF_{nDSM}} + \mathrm{Proj}(\mathrm{concat}(\mathbf{MF_{nDSM}}, \mathrm{Cro}_2(\mathbf{MF_{IRRG}}, \mathbf{MF_{nDSM}}))). \tag{10}$$

Finally, $\mathbf{OF_{IRRG}}$ and $\mathbf{OF_{nDSM}}$ are concatenated and embedded back into the original size $C \times H \times W$, producing the fused features as output

$$\textbf{Fused Feature} = \mathrm{Emb}(\mathrm{Concat}(\mathbf{OF_{IRRG}}, \mathbf{OF_{nDSM}})). \tag{11}$$

Through the cross-attention mechanism, the MFF module significantly improves feature fusion efficiency in the following aspects. First, it dynamically models the interaction relationships between modalities, enabling precise capture of critical region information. Second, by refining the computation of cross-modal weights, it maximizes the complementarity between modalities, avoiding redundancy caused by the simple aggregation of modal information in traditional methods. Additionally, the design of projection and context preservation ensures that the fused features enhance semantic expressiveness while maintaining the consistency of original modality information. Ultimately, the cross-attention mechanism makes the MFF process more efficient and semantically enriched, providing strong support for complex tasks.

IV. Experiments

In this section, we conduct comprehensive experiments on the Vaihingen and Potsdam datasets to evaluate the performance of our proposed FTransDeepLab. Our experiments include both qualitative and quantitative comparative analyses to thoroughly assess the framework's effectiveness. This section details the relevant data, settings, and results analysis.

A. Dataset

This article utilizes the publicly accessible German Vaihingen and Potsdam area datasets, disseminated by the ISPRS. The division of the dataset and each corresponding number are shown in Fig. 5. The remote sensing dataset categorizes the objects in the images into six major semantic classes: impervious surface, building, low vegetation, tree, car, and clutter.

The Vaihingen dataset consists of 33 images of varying sizes, approximately 500 × 2000 pixels each, cropped from larger scale aerial orthoimages. This dataset provides RGB, IRRG, DSM, nDSM, and label data. It includes three-band images and label data. According to the official dataset division, a total of 16 images are chosen for training: images 1,

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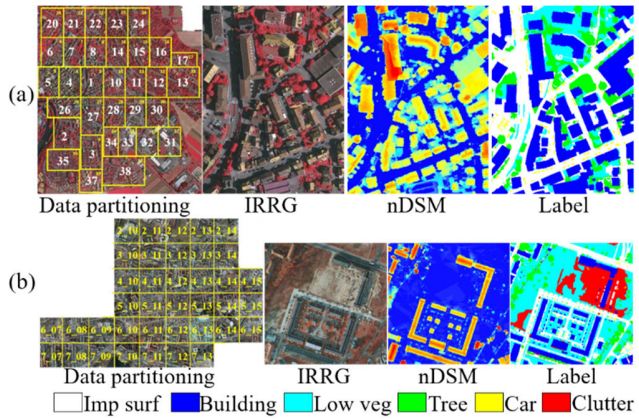


Fig. 5. (a) Vaihingen and (b) Potsdam datasets. (From left to right) Images show the data division map, IRRG, nDSM, and the labels. The dataset categorizes the objects in the images into six major semantic classes: impervious surface, building, low vegetation, tree, car, and clutter.

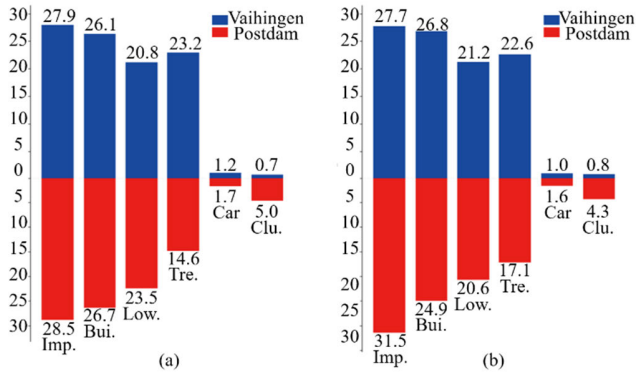


Fig. 6. Statistical distribution map of six categories. (a) Training set. (b) Testing set. In Vaihingen and Potsdam datasets, impervious surface and building cover the largest areas, followed by tree and low vegetation, while car and clutter are less prominent.

3, 5, 7, 11, 13, 15, 17, 21, 23, 26, 28, 30, 32, 34, and 37. The remaining 17 images are used for testing: images 2, 4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 27, 29, 31, 33, 35, and 38.

The Potsdam dataset comprises 38 aerial orthoimages, each with 6000 × 6000 pixels. This dataset provides RGB, IRRG, DSM, nDSM, and label data. For the experiments, a total of 24 images are used for training: images 2_10, 2_11, 2_12, 3_10, 3_11, 3_12, 4_10, 4_11, 4_12, 5_10, 5_11, 5_12, 6_7, 6_8, 6_9, 6_10, 6_11, 6_12, 6_7, 7_8, 7_9, 7_10, 7_11, and 7_12. The remaining 14 images are used for testing: images 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13, 5_14, 5_15, 6_13, 6_14, 6_15, and 7_13.

As shown in Fig. 6, the six-category distribution in the Vaihingen and Potsdam datasets are relatively consistent between the training (left) and testing (right) sets. In both datasets, impervious surface and building cover the largest area, reflecting their ubiquity and importance in urban areas. Tree and low vegetation follow, significant in natural landscapes and crucial for model recognition of natural environmental elements. In contrast, car and clutter are less prominent, possibly due to their smaller size, uneven distribution, or difficulty in recognition in aerial images. Although their impact on the overall model accuracy might be minor due to their lower overall semantic distribution weight, they are still important.

B. Experimental Setting

In this experiment, the hardware setup includes an NVIDIA A40 GPU, and the deep learning framework used is PyTorch. This study did not employ any external data augmentation methods outside the training process. Instead, random cropping, random flipping, and random rotation were used during training to enhance the model's robustness. For the Vaihingen and Potsdam datasets, 388 and 864 training samples of size 1024 × 1024 were generated, respectively. During the testing phase, no test-time augmentation (TTA) strategies were applied. Instead, direct inference with a size of 1024 × 1024 was performed, resulting in 113 and 504 test results for the two datasets, respectively. The encoder part of the model was initialized with Segformer (MiT-B2) to accelerate the training process and improve model performance. The AdamW optimizer was chosen for iterative training, with a batch size of 2 and an initial learning rate of 0.00006. The network was trained for 250 epochs until convergence. A linear learning rate warm-up strategy was applied for the first ten epochs, followed by a cosine annealing strategy to adjust the learning rate. The experiment employed the cross-entropy (CE) loss function to measure the difference between the model's predictions and the true labels. The loss function value is minimal when the model predictions perfectly match the true labels; otherwise, the value increases. The calculation formula is as follows:

$$\mathrm{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} y_{i,j}\log\left(p_{i,j}\right). \tag{12}$$

Here, N represents the total number of samples, C is the total number of categories, $y_{i,j}$ is the value of the jth category in the true label of sample i, and $p_{i,j}$ is the probability of the jth category in the model's prediction for sample i.

C. Evaluation Metrics

Semantic segmentation, a crucial task in CV, requires relevant metrics to validate the effectiveness of models. We use the ISPRS test set along with standard evaluation metrics, including the F1 score, overall accuracy (OA), and mean intersection over union (mIoU), all calculated based on the confusion matrix. In the confusion matrix, P represents the positive instances, and N represents the negative instances. True positive (TP) values are the elements on the diagonal of the confusion matrix, while TN represents the true negatives, FP represents the false positives, and FN represents the false negatives. The F1 score considers both precision and recall, serving as a metric to balance accuracy and completeness. OA represents the model's correct classification rate across all samples. Additionally, mIoU is used to evaluate the model's performance in segmentation tasks by measuring the overlap between the predicted and true segmentation areas for each class.

More specifically, we used the ground truth with eroded boundaries provided by ISPRS for calculations and computed the mF1 and mIoU for five foreground classes. Additionally, in the evaluation of OA, the clutter or background class was

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also taken into consideration. The formulas for F_1, OA, and mIoU are as follows:

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{13}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{14}$$

$$\text{mIoU} = \frac{1}{N} \sum_{i=1}^{N} \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i}. \tag{15}$$

Precision represents the proportion of actual positives among samples predicted as positive by the model, while recall represents the proportion of samples correctly predicted as positive among actual positives. The formulas are as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{16}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{17}$$

We evaluate the computational complexity of the proposed FTransDeepLab using the following evaluation metrics: the floating point operations (FLOPs) count, the number of model parameters, the memory footprint, and the frames per second (FPS). FLOPs are used to evaluate the model complexity whereas the number of model parameters and the memory footprint are used to evaluate the memory requirement. Finally, FPS is designed to assess the execution speed. Ideally, an efficient model should have smaller values in the first three metrics, but larger FPS value.

D. Performance Comparison

We benchmarked the performance of the proposed FTransDeepLab against several existing deep semantic models, including Unet [33], HRNet [62], Deeplabv3+ [34], DC-Swin [54], RS^{3}Mamba [63], BEDSN [58], ESANet [40], SFAFMANet [45], SGFNet [46], ASMFNet [64], FtransUnet [61], and MultiSenseSeg [47]. In our experiments, Unet, HRNet, Deeplabv3+, DC-Swin, BEDSN, RS^{3}Mamba, and BEDSN only considered the unimodal information, specifically visible images. These advanced single-modal methods illustrate the influence of nDSM data, demonstrating the advantages of multimodal approaches over single-modal ones. In contrast, the other methods accounted for multimodal data.

1) Performance Comparison on the Vaihingen Dataset: The comparison results of the proposed FTransDeepLab method with other unimodal and multimodal methods on the Vaihingen test set are shown in Table I. In terms of performance metrics, FTransDeepLab demonstrates significant superiority in remote sensing semantic segmentation tasks, achieving OA and mIoU scores of 93.16% and 84.91%, respectively. In the categories of impervious surface, building, low vegetation, tree, and car, FTransDeepLab achieves the highest F1 scores of 94.13%, 96.95%, 88.91%, 92.23%, and 86.35%, respectively. These high scores indicate that the proposed FTransDeepLab exhibits a high level of accuracy and robustness when handling complex geographical environments and different types of features. For underrepresented categories (such as cars) and easily confused categories (such as low vegetation and trees), FTransDeepLab also demonstrates a certain level further of adaptability. Table II presents the confusion matrix for the FTransDeepLab model on the Vaihingen validation set, clearly showcasing its performance. The rows represent the true categories, and the columns represent the model's predicted categories: 1) in the impervious surface category, the model performs excellently, achieving a recall of 0.94, and also demonstrates outstanding performance in the building category; 2) for the low vegetation category, the model achieves a recall of 0.90, with 0.07 of low vegetation misclassified as trees and 0.06 of trees misclassified as low vegetation; and 3) for the car category, despite its small sample size and inherent challenges, the model maintains an accuracy and a recall of 0.86, with 0.11 misclassified as impervious surfaces. Overall, FTransDeepLab demonstrates high accuracy and efficiency across all categories, particularly in the impervious surface and building categories. The primary misclassifications occur between low vegetation and trees. For underrepresented categories (such as cars), the model improves semantic distinction through multimodal fusion strategies and feature refinement, effectively mitigating the impact of category imbalance.

Fig. 7 presents some visualization examples from the Vaihingen. It can be observed that remote sensing images are more complex than natural images.

1) The complex shapes of large buildings, with varying heights and overlapping features, and their partial resemblance to low vegetation make it difficult to fully recognize building shapes, increasing the likelihood of misclassification as low vegetation, as shown in the first and second rows of Fig. 7.
2) The occlusion caused by building shadows, trees, and their shadows leads to incomplete or irregular building shapes, increasing the likelihood of misclassification as impervious surfaces, as shown in the third and fourth rows of Fig. 7.
3) Tree and low vegetation are interspersed, and due to their similar colors, misclassification is more likely to occur, as illustrated in the third and fourth rows of Fig. 7.
4) Small objects like cars are difficult to accurately label due to their small size, as seen in the fifth and sixth rows of Fig. 7. Clearly, the proposed FTransDeepLab can better address these challenges.

2) Performance Comparison on the Potsdam Dataset: The comparison results of the proposed FTransDeepLab method with other unimodal and multimodal methods on the Potsdam test set are shown in Table III. In terms of performance metrics, FTransDeepLab demonstrates significant superiority in remote sensing semantic segmentation tasks, achieving OA and mIoU scores of 91.63% and an mIoU of 86.97%, demonstrating the method's efficiency. Specifically, in the categories of impervious surface, building, low vegetation, tree, and car, FTransDeepLab achieves the highest F1 scores of 93.64%, 97.62%, 88.58%, 89.16%, and 95.55%, respectively. This result emphasizes the outstanding performance of FTransDeeplab in handling complex scenes. Table IV presents the confusion matrix of the FTransDeeplab model on the Potsdam validation set, clearly illustrating its performance.

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TABLE I
EXPERIMENTAL RESULTS ON THE VAIHINGEN TEST SET. WE PRESENT THE $F1$ OF FIVE FOREGROUND CLASSES AND THREE OVERALL PERFORMANCE METRICS. BOLD VALUES ARE THE BEST

Type	Method	Transformer	F1(%)						OA(%)	mIoU(%)
			Imp.	Bui.	Low.	Tre.	Car	Total		
Unimodal	Unet[33]	×	87.36	90.05	79.15	87.61	71.37	83.11	85.95	71.68
	Deeplabv3+[34]	×	91.35	93.84	83.26	88.75	77.19	86.88	89.23	77.28
	HRNet[62]	×	89.41	91.51	81.46	87.57	72.61	84.51	87.42	73.76
	DC-Swin[54]	√	92.77	95.05	84.92	89.79	78.56	88.22	90.26	80.89
	RS^{3}Mamba[63]	√	92.08	94.92	86.31	89.65	80.33	88.66	90.65	81.08
	BEDSN[58]	×	92.80	95.21	85.77	90.16	83.24	89.43	90.39	81.26
Multimodal	ESANet[40]	×	91.89	95.49	84.35	89.87	81.16	88.56	90.44	79.84
	SFAFMANet[45]	×	91.07	94.39	83.60	89.58	73.85	86.50	89.55	76.90
	SGFNet[46]	×	92.52	95.78	85.21	90.10	86.20	89.96	90.98	82.00
	FtransUnet[61]	√	93.01	96.37	88.36	89.53	**87.97**	91.25	92.08	83.06
	ASMFNet[64]	√	94.02	95.81	88.62	90.35	82.59	90.28	91.07	82.89
	MultiSenseSeg[47]	√	92.14	96.81	**89.00**	91.86	85.39	91.04	91.22	82.73
	FTransDeeplab(Ours)	√	**94.13**	**96.95**	88.91	**92.23**	86.35	91.71	93.16	**84.91**

TABLE II
CONFUSION MATRIX OF VAIHINGEN TEST SET. THE ITALICS REPRESENT A MISCLASSIFICATION RATE GREATER THAN 0.05. THE PRIMARY MISCLASSIFICATIONS OCCUR BETWEEN LOW VEGETATION AND TREES, AND CARS ARE OFTEN MISCLASSIFIED AS IMPERVIOUS SURFACE

	Imp.	Bui.	Low.	Tre.	Car	Clutter
Imp.	**0.94**	0.01	0.03	0.01	0.00	0.00
Bui.	0.02	**0.96**	0.01	0.00	0.00	0.00
Low.	0.03	0.00	**0.90**	*0.07*	0.00	0.00
Tre.	0.01	0.00	*0.06*	**0.93**	0.00	0.00
Car	*0.11*	0.00	0.02	0.00	**0.86**	0.00
Clutter	0.11	0.04	0.00	0.00	0.01	0.84
Precision	0.94	0.98	0.88	0.91	0.86	0.90
Recall	0.94	0.96	0.90	0.93	0.86	0.84
F1	0.94	0.97	0.89	0.92	0.86	0.87

It is evident that the results for Potsdam are similar to those for Vaihingen. This is reflected in: 1) the model's strong performance in identifying impermeable impervious surface and building, achieving recalls of 0.95 and 0.98, respectively, and 2) when observing the low vegetation category, the model's recall is 0.91, with 0.05 of low vegetation misclassified as tree. Correspondingly, 0.09 of trees are misclassified as low vegetation. In the car category, despite the smaller sample size, the model achieves a high recognition accuracy of 0.96. Overall, FTransDeepLab demonstrates stable performance across all categories, showcasing robustness in recognizing underrepresented categories, such as cars. For easily confused categories, such as low vegetation and trees, although some misclassifications still occur, the model has shown advantages in capturing detailed features and global semantics.

Fig. 8 presents some visualization examples from the Potsdam. It can be observed that Potsdam presents similar challenges to those in Vaihingen.

1) Building complexes and densely populated residential structures, although having neat boundaries, share similarities with the impervious surface in some areas, making them prone to confusion, as shown in the first and second rows of Fig. 8.
2) Trees and low vegetation are interspersed, and their similar colors make misclassification more likely, as illustrated in the third and fourth rows of Fig. 8.
3) Small objects, such as cars, are challenging to label accurately due to their small size, as shown in the fifth and sixth rows of Fig. 8.

V. DISCUSSION

A. Structure Analysis

To validate the effectiveness of our model, this study conducted a detailed comparison with FTrans + UNet, FTrans + Transformer, and FTrans + Deeplab (ours) on the Vaihingen datasets, as shown in Table V.

The results demonstrate that FTrans + DeepLab exhibits significant advantages in both segmentation performance and stability. On the Vaihingen dataset, FTrans + DeepLab achieved mIoU and OA scores of 84.91% and 93.16%, respectively, outperforming FTrans + UNet and FTrans + Transformer by over 2%. This highlights FTrans + DeepLab's superior ability to capture global features and handle fine details, making it more robust in complex scenarios. These results further confirm FTrans + DeepLab's stability across different datasets and scenarios. In terms of computational complexity, FTrans + Transformer had the lowest FLOPs

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TABLE III
EXPERIMENTAL RESULTS ON THE POTSDAM TEST SET. WE PRESENT THE $F1$ OF FIVE FOREGROUND CLASSES AND THREE OVERALL PERFORMANCE METRICS. BOLD VALUES ARE THE BEST

Type	Method	Transformer	F1(%)						OA(%)	mIoU(%)
			Imp.	Bui.	Low.	Tre.	Car	Total		
Unimodal	Unet[33]	×	87.05	88.74	80.24	81.81	91.97	85.96	83.58	75.64
	Deeplabv3+[34]	×	88.27	90.54	80.74	80.05	90.65	86.05	83.89	76.24
	HRNet[62]	×	88.87	91.08	80.94	79.52	92.09	86.50	84.40	76.58
	DC-Swin[54]	√	91.71	94.66	84.89	84.74	92.88	89.78	89.28	84.06
	RS^{3}Mamba[63]	√	90.52	93.22	87.95	88.62	91.68	90.40	89.67	85.06
	BEDSN[58]	×	90.67	94.89	87.13	87.84	92.35	90.58	89.79	84.48
Multimodal	ESANet[40]	×	91.41	97.26	85.07	85.50	94.51	90.75	89.11	83.43
	SFAFMANet[45]	×	90.30	96.25	83.45	83.44	91.40	88.97	87.47	80.48
	SGFNet[46]	×	92.69	96.88	87.03	87.53	94.87	91.80	90.33	85.08
	FtransUnet[61]	√	93.06	97.20	87.61	88.12	94.54	92.11	91.02	86.25
	ASMFNet[64]	√	**93.66**	95.93	86.88	86.12	94.51	91.42	90.14	85.11
	MultiSenseSeg[47]	√	93.55	96.52	86.84	87.93	94.91	91.95	90.86	85.72
	FTransDeeplab(Ours)	√	93.64	**97.62**	**88.58**	**89.16**	**95.55**	92.91	91.63	**86.97**

TABLE IV
CONFUSION MATRIX OF POTSDAM TEST SET. THE ITALICS REPRESENT A MISCLASSIFICATION RATE GREATER THAN 0.05. THE PRIMARY MISCLASSIFICATIONS OCCUR BETWEEN LOW VEGETATION AND TREES

	Imp.	Bui.	Low.	Tre.	Car	Clutter
Imp.	**0.95**	0.01	0.02	0.01	0.00	0.01
Bui.	0.01	**0.98**	0.00	0.00	0.00	0.01
Low.	0.03	0.00	**0.91**	*0.05*	0.00	0.01
Tre.	0.02	0.00	*0.09*	**0.88**	0.00	0.01
Car	0.01	0.00	0.01	0.01	**0.96**	0.01
Clutter	0.29	0.08	0.13	0.02	0.01	0.47
Precision	0.92	0.97	0.87	0.90	0.95	0.77
Recall	0.95	0.98	0.91	0.88	0.96	0.47
F1	0.94	0.98	0.89	0.89	0.96	0.58

TABLE V
PERFORMANCE COMPARISON OF FTRANS + UNET, FTRANS + TRANSFORMER, AND OUR MODEL ON VAIHINGEN DATASETS

Metrics	Structure		
	FTrans + Unet	FTrans + Transformer	FTrans + Deeplab
FLOPs (G)	14.27	**8.94**	12.94
Paramete (M)	66.56	**58.94**	69.86
Memory (MB)	811.92	**674.04**	755.3
Speed (FPS)	17.31	**18.26**	15.02
mF1 (%)	90.50	90.11	**91.71**
OA (%)	91.23	91.02	**93.16**
mIoU (%)	82.89	82.29	**84.91**

(8.94G) and parameter count (58.94M), while FTrans + DeepLab had FLOPs of 12.94G and parameter count of 69.86M, slightly higher than FTrans + Unet and FTrans + Transformer but still within a reasonable range. However, this minor computational cost resulted in a significant improvement in segmentation accuracy. Overall, FTrans + DeepLab achieves an excellent balance between segmentation performance and model complexity. Its ability to capture global features and effectively integrate multimodal information makes it highly competitive in semantic segmentation tasks, validating the effectiveness and advancement of its design.

B. Ablation Study

To validate the effectiveness of each proposed component in FTransDeepLab, ablation experiments were conducted by separately removing specific components while maintaining the framework.

As shown in Table VI, two ablation experiments were designed based on our proposed framework. In the first experiment, the MFR and MFF components were not introduced. In the second experiment, MFR was introduced to verify its effectiveness, while in the third experiment, both MFR and MFF were included to validate the effectiveness of the MFF component. Regarding baseline performance, the OA and mIoU of the base model were 88.87% and 73.68%, respectively. After introducing MFR, the OA and mIoU improved by 2.05% and 8.11%, respectively. With the addition of MFF, the OA and mIoU further increased by 2.24% and 3.12%. The performance improvement of the baseline model after incorporating MFR and MFF is particularly notable. The MFR module facilitates interaction between features extracted from one data source and features from another by learning weights, allowing the IRRG data to achieve better spatial structural understanding through correction with the nDSM, while the nDSM data gain richer texture and color information through correction with the IRRG. This enables

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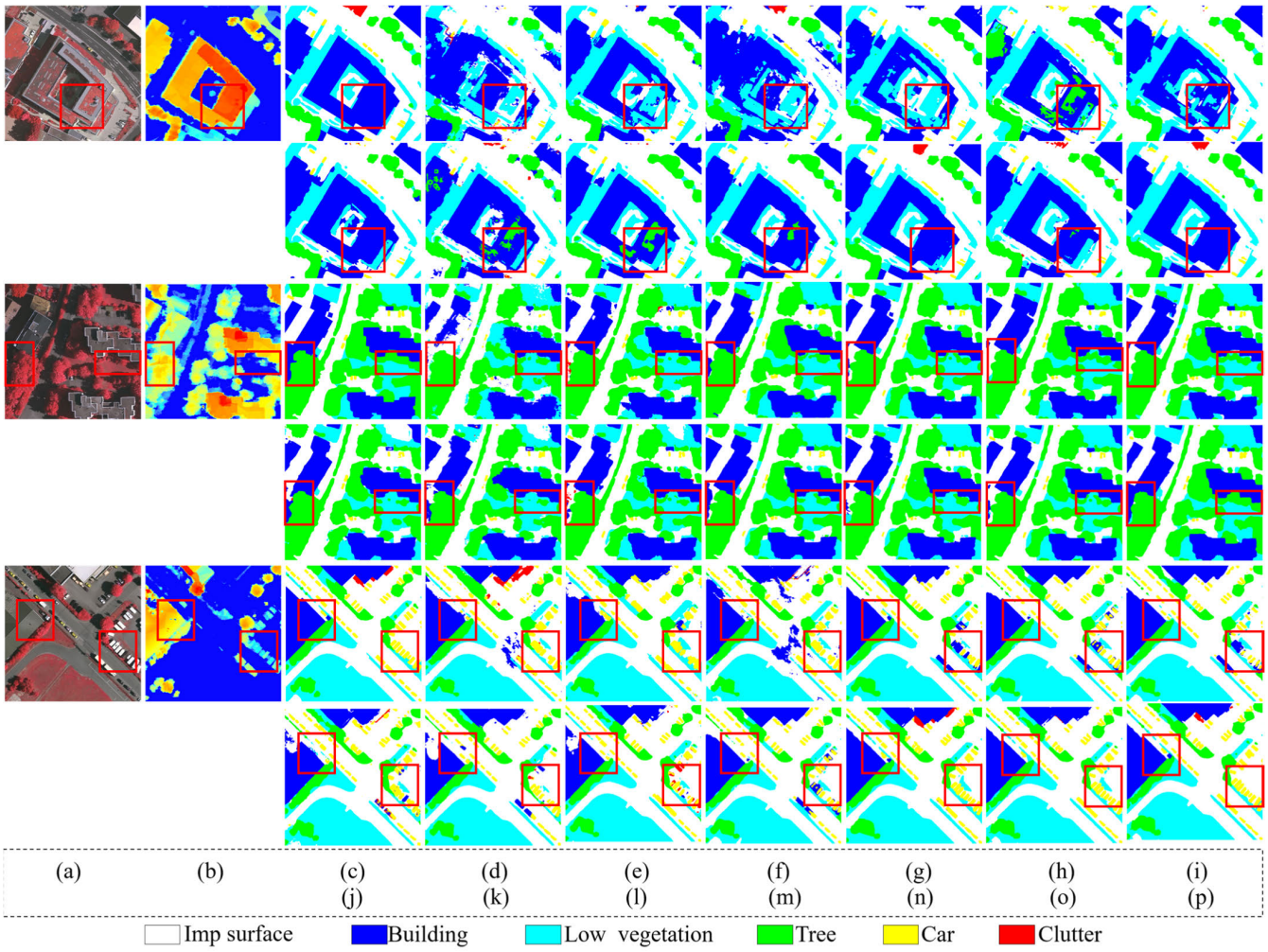


Fig. 7. Comparison of different models on the Vaihingen test set. (a) IRRG. (b) nDSM. (c) Ground truth. (d) UNet. (e) DeepLabV3+. (f) HRNet. (g) DC-Swin. (h) RS^{3}Mamba. (i) BEDSN. (j) ESANet. (k) SFAFMANet. (l) SGFNet. (m) FtransUnet. (n) ASMFNet. (o) MultiSenseSeg. (p) Ours. The segmentation results for different categories are marked in the following colors: impervious surface (white), building (blue), low vegetation (cyan), tree (green), car (yellow), and clutter (red). Red boxes are added to all subfigures to highlight the differences.

TABLE VI

ABLATION STUDY OF THE PROPOSED METHOD. THE RESULTS INDICATE THAT THE BASELINE MODEL SHOWS PARTICULARLY SIGNIFICANT PERFORMANCE IMPROVEMENTS AFTER THE INTRODUCTION OF MFR AND MFF

Modules		mF1	OA(%)	mIoU(%)
MFR	MFF			
		83.92	88.87	73.68
√		89.82	90.92	81.79
		(+5.90)	**(+2.05)**	**(+8.11)**
√	√	91.71	93.16	84.91
		(+1.89)	**(+2.24)**	**(+3.12)**

the model to integrate and emphasize these key pieces of information. Compared to simple linear operations, MFF optimizes the MFF process using a cross-attention mechanism. Fig. 9 qualitatively illustrates the advantages of the proposed approach.

To further validate the effectiveness of the proposed components in FTransDeepLab and their impact on multimodal data fusion, Fig. 10 intuitively illustrates the roles of the MFR and MFF modules at different feature levels and their influence on various categories. By processing and fusing the IRRG and nDSM feature maps, Fig. 10 clearly demonstrates the optimization effects of these modules during feature refinement and fusion. Specifically, Fig. 10(a) and (b) displays the original input feature maps of IRRG and nDSM, which serve as a baseline for comparison with the processed features, while Fig. 10(c) provides the ground truth as a reference standard for visual comparison. Fig. 10(d) and (e) visualizes the corrected IRRG and nDSM features at the low level after being processed by the MFR module, showcasing the ability of MFR to enhance detailed and spatial structure representation in low-level features. Fig. 10(f) presents the results of the low-level features fused by the MFF module, which significantly strengthens the collaborative expression of multimodal data through cross-modal feature interaction. Fig. 10(g) and (h) illustrates the corrected IRRG and nDSM features at the high level after being processed by the MFR module, while Fig. 10(i) demonstrates the fused high-level features obtained through the MFF module. Fig. 10(j)–(l) further visualizes the fused results in Fig. 10(i) by category, showing the pixel distributions for impervious surfaces, buildings, low vegetation, trees, and cars, respectively. These visualizations provide an

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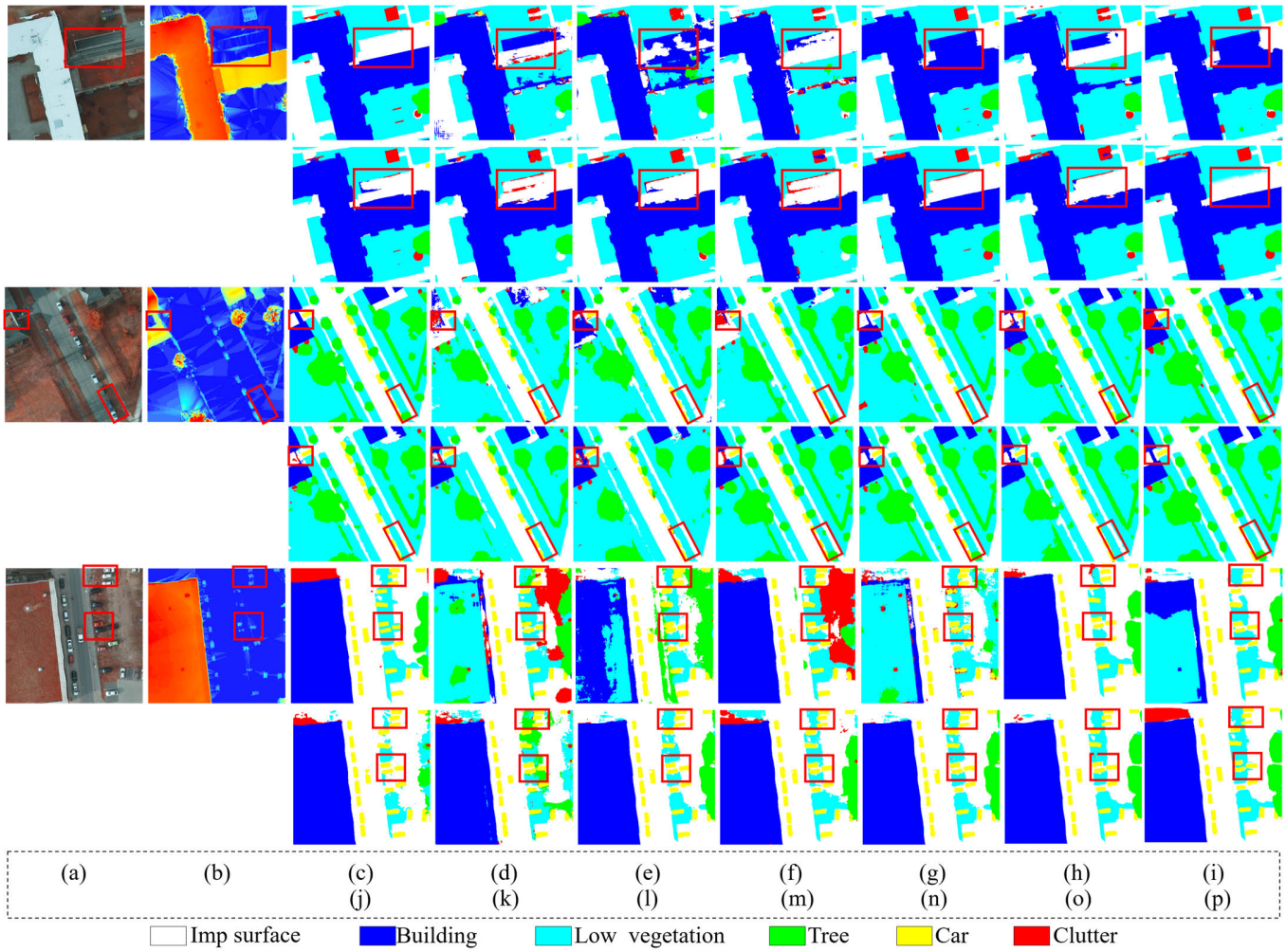


Fig. 8. Comparison of different models on the Potsdam test set. (a) IRRG. (b) nDSM. (c) Ground truth. (d) UNet. (e) DeepLabV3+. (f) HRNet. (g) DC-Swin. (h) RS^{3}Mamba. (i) BEDSN. (j) ESANet. (k) SFAFMANet. (l) SGFNet. (m) FtransUnet. (n) ASMFNet. (o) MultiSenseSeg. (p) Ours. The segmentation results for different categories are marked in the following colors: impervious surface (white), building (blue), low vegetation (cyan), tree (green), car (yellow), and clutter (red). Red boxes are added to all subfigures to highlight the differences.

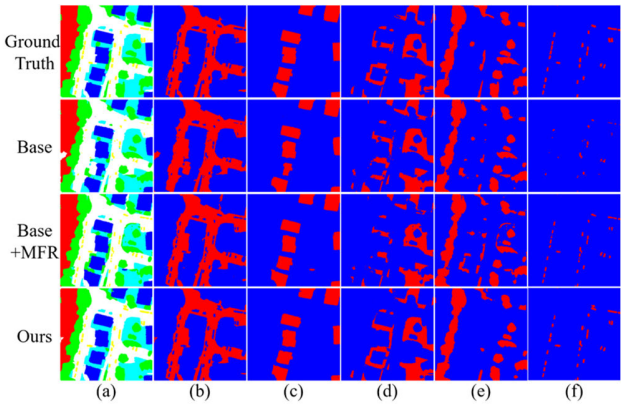


Fig. 9. Feature maps of ground truth, Base, Base + MFR, and ours. The larger the pixel value, the stronger the response. (a) Predictions for all categories. (b) Impervious surface. (c) Building. (d) Low vegetation. (e) Tree. (f) Car.

intuitive reflection of the model's ability to recognize different categories and its focus on key regions.

C. Resolution Analysis

To comprehensively evaluate the robustness and adaptability of the model under different image resolutions, this study designed a series of experiments to analyze the impact of resolution changes on the segmentation performance of different categories and overall metrics by controlling the variation in resolution. The primary objective of the experiments was to assess the stability and performance of the model as the resolution varied.

Specifically, the experimental data were cropped into three resolutions: 256 × 256, 512 × 512, and 1024 × 1024, to analyze the effect of resolution changes on the segmentation performance of different classes. The experimental results (Table VII) indicate that with increasing image resolution, the model consistently demonstrated improved performance across all categories and overall metrics.

The F1 score for the impervious surface category increased from 93.83% at 256 × 256 resolution to 94.13% at 1024 × 1024, showing only a small improvement. This suggests that this category primarily relies on overall regional features and is less sensitive to changes in resolution. The F1 score for the buildings category improved from 96.16% to 96.95%, as higher resolutions helped capture geometric shapes and boundary details more accurately, thereby reducing segmentation errors. Similarly, the F1 scores for low vegetation and

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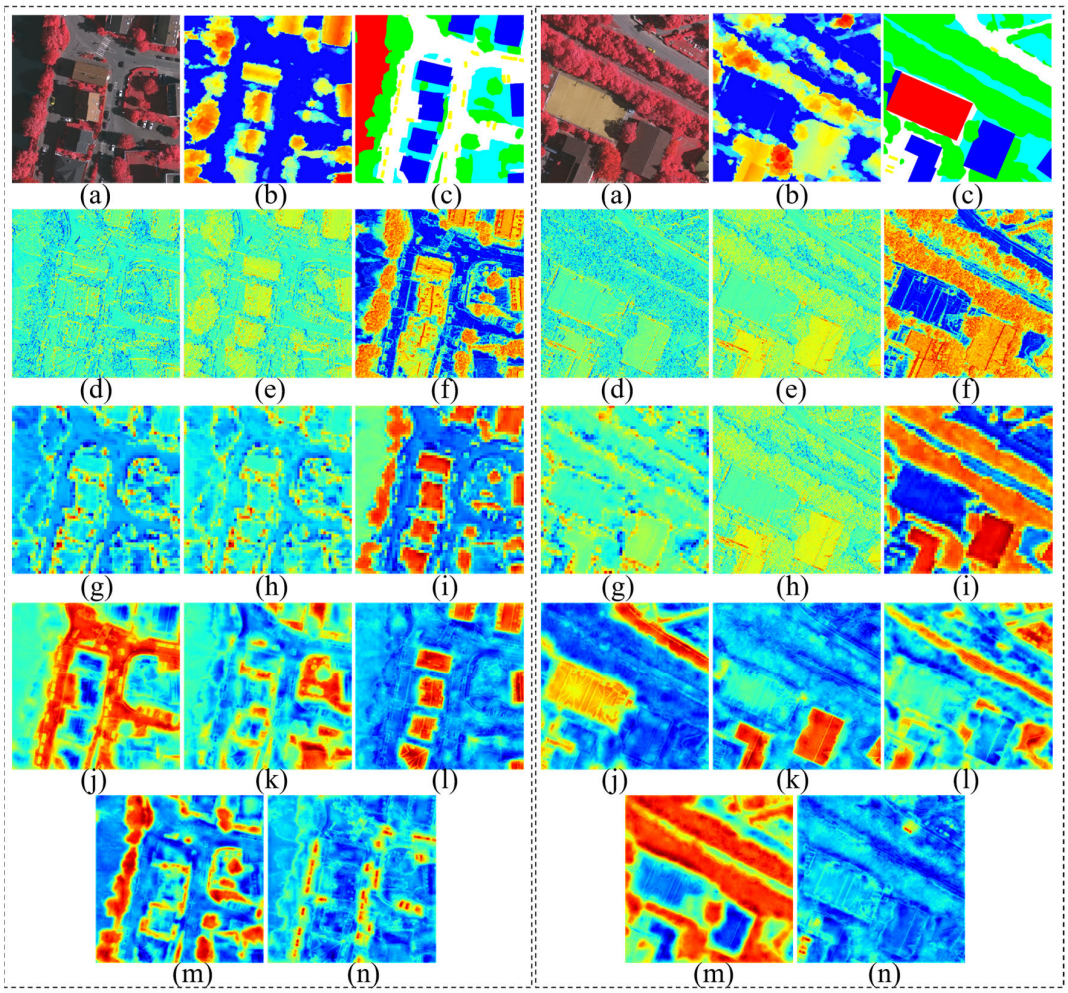


Fig. 10. Visualization of MFR and MFF as well as different classes. (a) IRRG. (b) nDSM. (c) Ground truth. (d) and (e) Visualizations of IRRG and nDSM processed by MFR in the low level. (f) Visualization of the fusion of IRRG and nDSM using MFF in the low level. (g) and (h) Visualizations of IRRG and nDSM processed by MFR in the high level. (i) Visualization of the fusion of IRRG and nDSM using MFF in the high level. (j)–(n) Pixels in (i) corresponding to impervious surface, building, low vegetation, tree, and car, respectively.

TABLE VII

SEGMENTATION PERFORMANCE OF DIFFERENT CLASSES UNDER DIFFERENT IMAGE RESOLUTIONS, INCLUDING THREE RESOLUTIONS: 256 × 256, 512 × 512, AND 1024 × 1024

Resolution		256*256	512*512	1024*1024
F1(%)	Imp.	93.83	94.08	94.13
	Bui.	96.16	96.49	96.95
	Low.	87.87	88.66	88.91
	Tre.	91.58	91.84	92.23
	Car	85.46	85.98	86.35
	Total	90.98	91.41	**91.71**
OA(%)		91.92	92.82	**93.16**
mIoU(%)		83.08	84.42	**84.91**

TABLE VIII

IMPACT OF HYPERPARAMETERS λ_C AND λ_S IN THE MFR MODULE ON MODEL PERFORMANCE. THREE CONFIGURATIONS WERE TESTED: $\lambda_C/\lambda_S = 0.25/0.75$, 0.5/0.5, AND 0.75/0.25

λ_C / λ_S		0.25 / 0.75	0.5/0.5 (ours)	0.75/0.25
F1(%)	Imp.	93.69	**94.13**	94.03
	Bui.	96.83	**96.95**	96.47
	Low.	88.62	88.91	**88.92**
	Tre.	91.92	92.23	**92.02**
	Car	**86.61**	86.35	86.02
	Total	91.53	**91.71**	91.49
OA(%)		92.86	**93.16**	92.81
mIoU(%)		84.79	**84.91**	84.27

trees improved, indicating that higher resolution enhances classification accuracy for these categories. The F1 score for cars increased from 85.46% to 86.35%, demonstrating that higher resolution benefits capturing detailed features and spatial positions of small objects like cars. These results show that higher resolution images improve the model's segmentation performance, particularly in capturing details and enhancing category differentiation. The qualitative results in Fig. 11 further illustrate the advantages of the proposed method.

D. Hyperparameter Analysis

To evaluate the impact of the hyperparameters λ_C and λ_S in the MFR on model performance, we conducted a sensitivity

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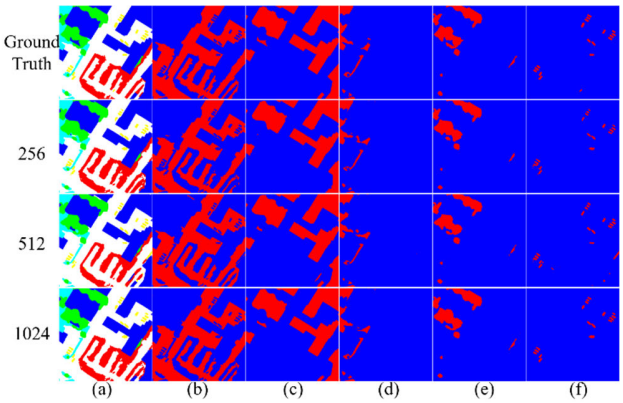


Fig. 11. Segmentation performance of different classes under different image resolutions, including three resolutions: 256 × 256, 512 × 512, and 1024 × 1024. (a) Predictions for all categories. (b) Impervious surfaces. (c) Building. (d) Low vegetation. (e) Tree. (f) Car.

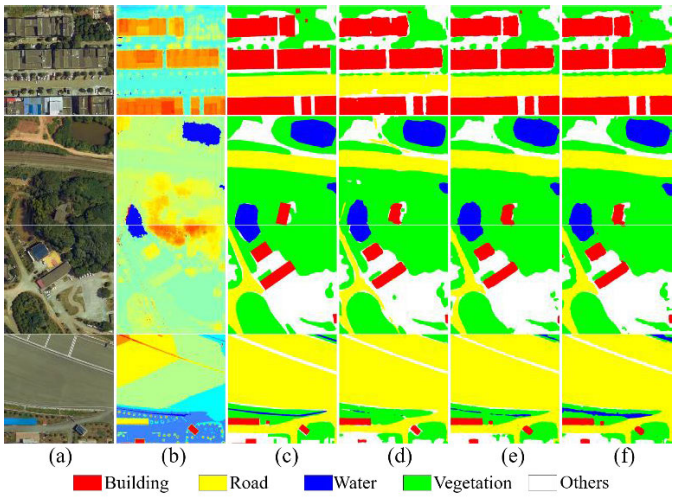


Fig. 12. Comparison of different models on the WHU-RRDSD test set. (a) RGB. (b) nDSM. (c) Ground truth. (d) FTransUnet. (e) MultiSenseSeg. (f) FTransDeepLab. The segmentation results for different categories are marked in the following colors: building (red), road (yellow), water (blue), vegetation (green), and others (white).

analysis on the ISPRS Vaihingen dataset, testing three configurations: $\lambda_C/\lambda_S = 0.25/0.75$, 0.5/0.5, and 0.75/0.25.

The experimental results (Table VIII) show that when $\lambda_C = 0.25$ and $\lambda_S = 0.75$, the model performs well, particularly in capturing boundary and detailed features, with advantages in F1 scores for the building and car categories. However, its performance in the low vegetation and tree categories is slightly weaker. In contrast, the balanced configuration of $\lambda_C = 0.5$ and $\lambda_S = 0.5$ achieves the best overall performance, effectively capturing global semantic information while maintaining balanced performance on local features. When $\lambda_C = 0.75$ and $\lambda_S = 0.25$, the model shows a preference for global feature representation, with improved performance in the low vegetation and tree categories but slightly decreased performance in the building and car categories. In summary, different hyperparameter configurations highlight varying emphases on channel and spatial attentions. The configuration of $\lambda_C = 0.25$ and $\lambda_S = 0.75$ is better suited for tasks emphasizing local features, while $\lambda_C = 0.75$ and $\lambda_S = 0.25$ is more sensitive to global features. The balanced configuration of $\lambda_C = 0.5$ and $\lambda_S = 0.5$ achieves optimal overall performance.

E. Model Complexity Analysis

Table IX presents the complexity analysis results for all the comparison methods considered in this article. It can be observed that the proposed FTransDeepLab has relatively lower values for FLOPs and memory usage. In terms of the number of parameters, the multimodal methods SGFNet and FTransUnet have 125.25M and 160.88M parameters, respectively, indicating that their network structures are more complex and have higher computational costs. In contrast, our model only has 69.86M parameters. Additionally, FTransDeepLab shows significant advantages in segmentation performance among multimodal methods. In terms of the key metric mIoU, FTransDeepLab achieves 84.91%, which is 2.92% higher than SGFNet and 1.85% higher than FTransUnet. Notably, despite the significant reduction in model complexity, FTransDeepLab still provides excellent segmentation performance in multimodal scenarios, particularly in capturing details and identifying boundaries in complex scenes. FTransDeepLab achieves an inference time of 66.66 ms and shows lower GPU memory usage (108.75M) on our hardware setup (NVIDIA A40 GPU), further demonstrating its efficient computational performance and low memory consumption. These results validate the effectiveness of the model structure design and MFF strategy of FTransDeepLab, highlighting the effective balance between performance improvement and complexity optimization in multimodal methods.

F. Model Generalization Ability Analysis

Considering the diversity of remote sensing images, the model's generalization ability is crucial. In this study, the WHU-RRDSD dataset is chosen to further evaluate the performance of FTransDeepLab in different scenarios.

The WHU-RRDSD dataset [68] includes five land cover classes (building, road, water, vegetation, and others), covering the area along the Changgan Railway. The original images are 6060 × 6060 pixels, which are cropped to 1024 × 1024 pixels, with a total of 612 images. Among them, the training set consists of 367 images, and the test set consists of 245 images. In the experiments, models such as FTransUnet, MultiSenseSeg, and FTransDeepLab are compared, and the performance of each model is evaluated using F1 scores for four foreground classes, as well as three overall performance metrics, including OA and mIoU. The experimental results (Table X) show that FTransDeepLab excels in F1 score, OA, and mIoU, especially demonstrating significant advantages in the segmentation of building, road, water, and vegetation classes. This indicates that FTransDeepLab has strong generalization ability on this dataset and can effectively handle segmentation tasks for different land covers. Therefore, FTransDeepLab maintains high accuracy and robustness in the diversity of remote sensing images. Fig. 12 qualitatively demonstrates the model's segmentation results.

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TABLE IX
COMPUTATIONAL COMPLEXITY ANALYSIS MEASURED BY A 256 × 256 IMAGE ON A SINGLE NVIDIA GEFORCE NVIDIA A40 GPU. MIOU VALUES ARE THE RESULTS ON THE VAIHINGEN DATASET. BOLD VALUES ARE THE BEST

Type	Method	FLOPs (G)	Parameter (M)	Memory (MB)	Speed (FPS)	Inference Time(ms)	GPU Usage (M)	MIoU(%)
Unimodal	Unet[33]	6.52	31.02	396.58	47.55	21.05	57.14	71.68
	Deeplabv3+[34]	7.49	39.06	450.15	32.30	30.96	65.74	77.28
	HRNet[62]	36.97	112.66	890.47	18.58	53.74	116.12	73.76
	DC-Swin[54]	34.44	96.02	870.65	13.94	71.92	108.13	80.89
	BEDSN[58]	10.95	**21.27**	565.41	48.63	24.62	73.73	81.26
Multimodal	ESANet[40]	**5.21**	23.92	**377.57**	**49.45**	**20.22**	**50.38**	79.84
	SFAFMANet[45]	18.26	86.89	1371.34	15.52	64.40	208.28	76.90
	SGFNet[46]	30.85	125.25	1441.74	10.15	98.52	215.65	81.99
	FtransUnet[61]	45.68	160.88	1653.24	8.81	113.63	237.77	83.06
	MultiSenseSeg[47]	10.24	58.66	595.32	21.83	45.80	87.24	82.73
	FTransDeeplab (Ours)	12.94	69.86	755.30	15.02	66.66	108.75	**84.91**

TABLE X
PERFORMANCE OF FTRANSUNET, ASMFNET, AND FTRANSDEEPLAB ON THE WHU-RRDSD TEST SET. WE PRESENT THE F1 SCORES FOR FOUR FOREGROUND CLASSES AND THREE OVERALL PERFORMANCE METRICS. BOLD VALUES INDICATE THE BEST RESULTS

Method		Ftrans-Unet	Multi-SenseSeg	FTrans-Deeplab
F1(%)	Building	90.90	**91.35**	91.10
	Road	90.29	90.93	**91.81**
	Water	97.02	**97.98**	97.90
	Vegetation	93.05	93.22	**93.29**
	Total	92.05	93.37	93.53
OA(%)		94.71	95.02	95.32
mIoU(%)		85.65	85.96	**86.79**

VI. CONCLUSION

To accurately capture semantic information in remote sensing images, this study proposes FTransDeepLab, a remote sensing semantic segmentation model based on DeepLabv3+, which integrates multimodal fusion and Transformer architecture, aiming to enhance the semantic parsing capability of high-resolution remote sensing imagery. Compared to single-modal methods, this model fully leverages the complementary information from multimodal data and introduces a multiscale Segformer structure in the encoder to improve long-range dependence modeling and multiscale feature extraction. Additionally, we designed two modules, MFR and MFF, to optimize multimodal feature integration. MFR enhances target region analysis through feature reconstruction, particularly excelling in scenes with rich textures or complex geometric structures. MFF dynamically adjusts the contribution of different modalities, refining the joint representation of multimodal data and further improving segmentation accuracy. The experimental results demonstrate that FTransDeepLab outperforms the state-of-the-art semantic segmentation methods on the ISPRS Vaihingen and Potsdam datasets. Moreover, while maintaining high segmentation performance, it significantly reduces computational complexity, enhancing its practical applicability. However, the method still has certain limitations, such as its strong reliance on high-quality multimodal data, which constrains its generalization in data-limited scenarios. Therefore, future work can focus on improving adaptability to low-quality data to further expand the model's applicability.

ACKNOWLEDGMENT

The authors would like to express their gratitude to Prof. Hu, Dr. Zheng, and Dr. Wang.

REFERENCES

[1] S. Hao, Y. Zhou, and Y. Guo, "A brief survey on semantic segmentation with deep learning," *Neurocomputing*, vol. 406, pp. 302–321, Sep. 2020.
[2] Q. Weng, H. Chen, H. Chen, W. Guo, and Z. Mao, "A multisensor data fusion model for semantic segmentation in aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[3] S. Chen, W. Shi, M. Zhou, M. Zhang, and P. Chen, "Automatic building extraction via adaptive iterative segmentation with LiDAR data and high spatial resolution imagery fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2081–2095, 2020.
[4] X. Yang et al., "An attention-fused network for semantic segmentation of very-high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 238–262, Jul. 2021.
[5] S. Arora, J. Acharya, A. Verma, and P. K. Panigrahi, "Multilevel thresholding for image segmentation through a fast statistical recursive algorithm," *Pattern Recognit. Lett.*, vol. 29, no. 2, pp. 119–125, Jan. 2008.
[6] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[7] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

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[8] K. Islam, "Recent advances in vision transformer: A survey and outlook of recent work," 2022, *arXiv:2203.01536*.

[9] K. Han et al., "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, Jan. 2022.

[10] Y. Gao, M. Zhou, and D. N. Metaxas, "UTNet: A hybrid transformer architecture for medical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, Strasbourg, France. Cham, Switzerland: Springer, Oct. 2021, pp. 61–71.

[11] Y. Jin, D. Han, and H. Ko, "TrSeg: Transformer for semantic segmentation," *Pattern Recognit. Lett.*, vol. 148, pp. 29–35, Aug. 2021.

[12] Z. Xu, W. Zhang, T. Zhang, Z. Yang, and J. Li, "Efficient transformer for remote sensing image segmentation," *Remote Sens.*, vol. 13, no. 18, p. 3585, Sep. 2021.

[13] R. Azad, M. T. Al-Antary, M. Heidari, and D. Merhof, "TransNorm: Transformer provides a strong spatial normalization mechanism for a deep segmentation model," *IEEE Access*, vol. 10, pp. 108205–108215, 2022.

[14] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[15] H. Cao et al., "Swin-UNet: UNet-like pure transformer for medical image segmentation," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 205–218.

[16] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.

[17] C. Han, C. Wu, H. Guo, M. Hu, and H. Chen, "HANet: A hierarchical attention network for change detection with bitemporal very-high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3867–3878, 2023.

[18] J. Li et al., "Deep learning in multimodal remote sensing data fusion: A comprehensive review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102926.

[19] Z. Xiong, Y. Yuan, and Q. Wang, "ASK: Adaptively selecting key local features for RGB-D scene recognition," *IEEE Trans. Image Process.*, vol. 30, pp. 2722–2733, 2021.

[20] H. Qin, W. Zhou, Y. Yao, and W. Wang, "Individual tree segmentation and tree species classification in subtropical broadleaf forests using UAV-based LiDAR, hyperspectral, and ultrahigh-resolution RGB data," *Remote Sens. Environ.*, vol. 280, Oct. 2022, Art. no. 113143.

[21] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[22] X. Wang, K. Li, and A. Chehri, "Multi-sensor fusion technology for 3D object detection in autonomous driving: A review," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 2, pp. 1148–1165, Feb. 2024.

[23] X. Li, C. Wen, L. Wang, and Y. Fang, "Geometry-aware segmentation of remote sensing images via joint height estimation," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[24] J. Zhang, H. Liu, K. Yang, X. Hu, R. Liu, and R. Stiefelhagen, "CMX: Cross-modal fusion for RGB-X semantic segmentation with transformers," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 12, pp. 14679–14694, Dec. 2023.

[25] J. Cao, H. Leng, D. Lischinski, D. Cohen-Or, C. Tu, and Y. Li, "ShapeConv: Shape-aware convolutional layer for indoor RGB-D semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7068–7077.

[26] L.-Z. Chen, Z. Lin, Z. Wang, Y.-L. Yang, and M.-M. Cheng, "Spatial information guided convolution for real-time RGBD semantic segmentation," *IEEE Trans. Image Process.*, vol. 30, pp. 2313–2324, 2021.

[27] G. Xiao, D. P. Bavirisetti, G. Liu, and X. Zhang, "Decision-level image fusion," in *Image Fusion*. Cham, Switzerland: Springer, 2020, pp. 149–170.

[28] J. Chen et al., "Ship target detection algorithm based on decision-level fusion of visible and SAR images," *IEEE J. Miniaturization Air Space Syst.*, vol. 4, no. 3, pp. 242–249, Mar. 2023.

[29] S. Feng et al., "CPFNet: Context pyramid fusion network for medical image segmentation," *IEEE Trans. Med. Imag.*, vol. 39, no. 10, pp. 3008–3018, Oct. 2020.

[30] Q. Zhang, S. Zhao, Y. Luo, D. Zhang, N. Huang, and J. Han, "ABMDRNet: Adaptive-weighted bi-directional modality difference reduction network for RGB-T semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2633–2642.

[31] S. Dong and Z. Chen, "A multi-level feature fusion network for remote sensing image segmentation," *Sensors*, vol. 21, no. 4, p. 1267, Feb. 2021.

[32] K. Lu, J. Cheng, H. Li, and T. Ouyang, "MFAFNet: A lightweight and efficient network with multi-level feature adaptive fusion for real-time semantic segmentation," *Sensors*, vol. 23, no. 14, p. 6382, Jul. 2023.

[33] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Med. Image Comput. Comput.-Assist. Intervent.-MICCAI*, Munich, Germany. Cham, Switzerland: Springer, Oct. 2015, pp. 234–241.

[34] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2017.

[35] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.

[36] J. Gupta, "Learning rich features from RGB-D images for object detection and segmentation," in *Proc. Eur. Conf. Comput. Vis.*, Zurich, Switzerland. Cham, Switzerland: Springer, Sep. 2014, pp. 345–360.

[37] C. Hazirbas, L. Ma, C. Domokos, and D. Cremers, "FuseNet: Incorporating depth into semantic segmentation via fusion-based CNN architecture," in *Proc. Asian Conf. Comput. Vis.*, Taipei, Taiwan. Cham, Switzerland: Springer, Nov. 2016, pp. 213–228.

[38] X. Hu, K. Yang, L. Fei, and K. Wang, "ACNET: Attention based network to exploit complementary features for RGBD semantic segmentation," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 1440–1444.

[39] S. Chen, X. Zhu, W. Liu, X. He, and J. Liu, "Global-local propagation network for RGB-D semantic segmentation," 2021, *arXiv:2101.10801*.

[40] J. Zhou, S. Qian, Z. Yan, J. Zhao, and H. Wen, "ESA-net: A network with efficient spatial attention for smoky vehicle detection," in *Proc. IEEE Int. Instrum. Meas. Technol. Conf. (I2MTC)*, May 2021, pp. 1–6.

[41] Y. Wang, X. Chen, L. Cao, W. Huang, F. Sun, and Y. Wang, "Multimodal token fusion for vision transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 12176–12185.

[42] Z. Wan et al., "Sigma: Siamese mamba network for multi-modal semantic segmentation," 2024, *arXiv:2404.04256*.

[43] Z. Ma, Z. Jiang, and H. Zhang, "Hyperspectral image classification using feature fusion hypergraph convolution neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517314.

[44] S. He, H. Yang, X. Zhang, and X. Li, "MFTransNet: A multi-modal fusion with CNN-transformer network for semantic segmentation of HSR remote sensing images," *Mathematics*, vol. 11, no. 3, p. 722, Feb. 2023.

[45] X. He, M. Wang, T. Liu, L. Zhao, and Y. Yue, "SFAF-MA: Spatial feature aggregation and fusion with modality adaptation for RGB-thermal semantic segmentation," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–10, 2023.

[46] Y. Wang, G. Li, and Z. Liu, "SGFNet: Semantic-guided fusion network for RGB-thermal semantic segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 12, pp. 7737–7748, Dec. 2023.

[47] Q. Wang, W. Chen, Z. Huang, H. Tang, and L. Yang, "MultiSenseSeg: A cost-effective unified multimodal semantic segmentation model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4703724.

[48] P. Tokarczyk, J. D. Wegner, S. Walk, and K. Schindler, "Features, color spaces, and boosting: New insights on semantic classification of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 280–295, Jan. 2015.

[49] M. Dalla Mura, J. A. Benediktsson, B. Waske, and L. Bruzzone, "Morphological attribute profiles for the analysis of very high resolution images," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 10, pp. 3747–3762, Oct. 2010.

[50] K. Yue, L. Yang, R. Li, W. Hu, F. Zhang, and W. Li, "TreeUNet: Adaptive tree convolutional neural networks for subdecimeter aerial image segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 156, pp. 1–13, Oct. 2019.

[51] F. I. Diakogiannis, F. Waldner, P. Caccetta, and C. Wu, "ResUNet-a: A deep learning framework for semantic segmentation of remotely sensed data," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 94–114, Apr. 2020.

[52] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4095–4104.

[53] A. Ma, J. Wang, Y. Zhong, and Z. Zheng, "FactSeg: Foreground activation-driven small object semantic segmentation in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2021.

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[54] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[55] X. Kang, Y. Hong, P. Duan, and S. Li, "Fusion of hierarchical class graphs for remote sensing semantic segmentation," *Inf. Fusion*, vol. 109, Sep. 2024, Art. no. 102409.
[56] C. Peng, K. Zhang, Y. Ma, and J. Ma, "Cross fusion net: A fast semantic segmentation network for small-scale semantic information capturing in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601313.
[57] J. Jin, W. Zhou, R. Yang, L. Ye, and L. Yu, "Edge detection guide network for semantic segmentation of remote-sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[58] X. Li, L. Xie, C. Wang, J. Miao, H. Shen, and L. Zhang, "Boundary-enhanced dual-stream network for semantic segmentation of high-resolution remote sensing images," *GIScience Remote Sens.*, vol. 61, no. 1, Dec. 2024, Art. no. 2356355.
[59] T. Xiao, Y. Liu, Y. Huang, M. Li, and G. Yang, "Enhancing multiscale representations with transformer for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605116.
[60] Q. Zhao, J. Liu, Y. Li, and H. Zhang, "Semantic segmentation with attention mechanism for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5403913.
[61] X. Ma, X. Zhang, M.-O. Pun, and M. Liu, "A multilevel multimodal fusion transformer for remote sensing semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5403215.
[62] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2020.
[63] X. Ma, X. Zhang, and M.-O. Pun, "RS^{3}Mamba: Visual state space model for remote sensing image semantic segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[64] X. Ma, X. Xu, X. Zhang, and M.-O. Pun, "Adjacent-scale multimodal fusion networks for semantic segmentation of remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 20116–20128, 2024.
[65] X. Li, M. Ding, and A. Pižurica, "Deep feature fusion via two-stream convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2615–2629, Apr. 2020.
[66] J. Yang, L. Xiao, Y.-Q. Zhao, and J. C. Chan, "Unsupervised deep tensor network for hyperspectral–multispectral image fusion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 9, pp. 13017–13031, Sep. 2024.
[67] J. Xue, Y.-Q. Zhao, T. Wu, and J. Cheung-Wai Chan, "Tensor convolution-like low-rank dictionary for high-dimensional image representation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 12, pp. 13257–13270, Dec. 2024.
[68] H. Feng et al., "Method for constructing a remote sensing image sample library for intelligent interpretation of railway design control elements," *Railway Standard Des.*, vol. 69, no. 3, pp. 48–57, Mar. 2025.



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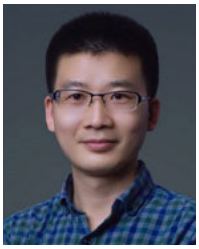
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